IRCA Exposure and Composition of Admissions

A descriptive exposure-gradient analysis, 1982–2000

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February 2026

Executive Summary

This project documents whether higher exposure to the 1986 Immigration Reform and Control Act (IRCA) is associated with changes in the occupational composition of immigrants admitted to the United States from Latin American origin countries between 1982 and 2000. The analysis uses a country–year panel (19 origins) and measures exposure intensity as IRCA legalizations during 1989–1992 relative to pre-IRCA admissions (1982–1985).

Main findings:

- Higher IRCA exposure is associated with a decline in the blue-collar share among recorded occupations during 1989–1992: -3.6 pp per one-unit exposure (clustered SE 0.7 pp).
- The association is stable across common sensitivity checks (country trends, alternative weights, balanced panel restriction, leave-one-out, excluding Mexico, and window restrictions).
- Timing diagnostics are mixed: a linear pre-period trend test and pre-period placebo "fake-post" tests are small and imprecise, but an event-study joint test for pre-1988 coefficients rejects strongly (Wald $F(6, 298) = 13.54$, $p = 1.38 \times 10^{-13}$). I therefore treat dynamic estimates as descriptive timing patterns and rely primarily on robustness and aggregated contrasts for interpretation.
- Results are sensitive to outcome construction: when "no occupation" is included in the denominator ("full denominator"), the blue-share association attenuates toward zero (≈ -0.5 pp; not statistically significant).
- A decomposition in levels and logs suggests the share decline reflects faster growth in denominators than the blue-collar numerator: blue-collar counts rise with exposure (+9,683), but total/known occupation counts rise more (+39k to +43k).

All results are reproducible from raw Excel inputs through a single `run.R` pipeline that exports tables, figures, logs, and a clean panel dataset.

Stakeholder takeaway.

For monitoring and reporting, these results suggest that large legalization/processing surges can coincide with changes in recorded occupational composition—sometimes because denominators

change (including shifts into "no occupation"), not because a category collapses in levels.

Recommendation: track three metrics together during surge windows: (1) blue-collar share among known occupations, (2) no-occupation share, and (3) underlying numerator/denominator counts. This helps distinguish genuine compositional change from measurement/recording shifts and supports capacity planning (processing staff, classification QA, and service delivery).

Research Question

Do origin countries with higher IRCA legalization intensity display larger changes in the occupational composition of admissions during the 1989–1992 legalization surge window?

Rather than a binary treated/control design, I use cross-country variation in exposure intensity:

$$\text{Exposure} = \frac{\text{IRCA legalizations (1989–1992)}}{\text{Pre-IRCA admissions (1982–1985)}}.$$

Primary outcomes are occupational shares: blue-collar, agriculture, manual, services, white-collar, and "no occupation" (missing/unknown occupation category). I also analyze class-of-admission shares and selected demographic composition measures as exploratory outcomes.

Data and Pipeline

Data come from INS Statistical Yearbooks (now OHSS), organized into four raw inputs:

- occupation by origin–year
- class of admission by origin–year
- gender/age by origin–year
- adjustments of status vs new arrivals by origin–year

Pipeline features:

- Automatic detection of raw Excel files
- Multi-sheet ingestion using a dynamic year parser
- Construction of country–year panel (1982–2000)
- Creation of derived shares and treatment intensity
- Missingness diagnostics
- Reproducible export of all tables (.tex) and figures (.png)

Final dataset:

- 361 country–year observations, 19 origins, 1982–2000.

Data QA gates: uniqueness of country–year keys after merges, non-negative counts, and share variables constrained to [0,1]. The run aborts if these checks fail.

Table 1: Missingness by variable at the country–year level.

Variable	Obs.	Missing	Total	Share missing (%)
Male share	133	228	361	63.2
Family share	352	9	361	2.5
Employment share	352	9	361	2.5
Humanitarian share	352	9	361	2.5
Other share	352	9	361	2.5
Adjustment share	352	9	361	2.5
Blue share	353	8	361	2.2
Agriculture share	353	8	361	2.2
Manual share	353	8	361	2.2
Services share	353	8	361	2.2
White share	353	8	361	2.2
No-occupation share	353	8	361	2.2

Table 2: Coverage by country for key variables.

Country	Years total	Years covered (key vars)	Coverage ratio
Argentina	19	19	1.000
Bolivia	19	19	1.000
Brazil	19	19	1.000
Chile	19	19	1.000
Colombia	19	19	1.000
Costa Rica	19	19	1.000
Cuba	19	19	1.000
Dominican Republic	19	19	1.000
Ecuador	19	19	1.000
El Salvador	19	19	1.000
Guatemala	19	19	1.000
Honduras	19	19	1.000
Mexico	19	19	1.000
Nicaragua	19	19	1.000
Panama	19	19	1.000
Peru	19	19	1.000
Venezuela	19	19	1.000
Uruguay	19	15	0.789
Paraguay	19	13	0.684

How to Run

1. Place the raw Excel inputs in `data_raw/`.
2. "Run the pipeline: `Rscript run.R`"
3. Render this report

Project Structure

project/ data_raw/ # raw Excel inputs data_clean/ panel.rds # cleaned country-year panel outputs/ tables/ # exported LaTeX tables figures/ # exported figures logs/ # run log + sessionInfo + summary run.R # pipeline script paper.Rmd # report

Modeling Approach (Descriptive Association)

I estimate descriptive association models that summarize how outcomes co-move with exposure during 1989–1992, controlling for country fixed differences and common year shocks. This is useful for documenting exposure gradients and timing patterns; it is not a randomized design, so results are interpreted as associations.

$$Y_{c,t} = \beta\,(\text{Exposure}_c \times \text{Post}_t) + \alpha_c + \delta_t + \varepsilon_{c,t}.$$

Where:

- α_c = country fixed effects
- δ_t = year fixed effects
- Post = 1989–1992
- SE clustered at country level

Net specifications include AdjShare to absorb mechanical processing variation during IRCA.

Main Results

Tables 3–4 show the Total specification (no adj_share). Tables 5–6 show the Net specification (includes adj_share).

Table 3: Baseline TWFE (Total) — Panel A: occupation shares (Blue–White).

	Blue	Agri	Manual	Services	White
IRCA intensity × Post	−0.025***	0.029***	0.004	−0.007**	0.004
	(0.005)	(0.004)	(0.005)	(0.003)	(0.005)
Num.Obs.	353	353	353	353	353
Mean of Y	0.342	0.050	0.392	0.224	0.384

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp changes per one-unit exposure. Total specs exclude adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 4: Baseline TWFE (Total) — Panel B: remaining outcomes (No-Occ + class shares).

	No-Occ	Family	Employment	Humanitarian	Other
IRCA intensity × Post	−0.035***	−0.052***	−0.021*	−0.004***	−0.002
	(0.007)	(0.004)	(0.011)	(0.001)	(0.001)
Num.Obs.	353	352	352	352	352
Mean of Y	0.620	0.542	0.301	0.067	0.013

* p <0.1, ** p <0.05, *** p <0.01

Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp changes per one-unit exposure. Total specs exclude adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 5: Baseline TWFE (Net) — Panel A: occupation shares (Blue–White).

	Blue	Agri	Manual	Services	White
IRCA intensity × Post	−0.036***	0.029***	−0.007	−0.006*	0.013*
	(0.007)	(0.004)	(0.006)	(0.003)	(0.006)
Adjustment share	0.169*	−0.006	0.164*	−0.019	−0.144
	(0.085)	(0.027)	(0.082)	(0.032)	(0.093)
Num.Obs.	351	351	351	351	351
Mean of Y	0.342	0.050	0.392	0.224	0.384

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp per one-unit exposure. Net specs include adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 6: Baseline TWFE (Net) — Panel B: remaining outcomes (No-Occ + class shares).

	No-Occ	Family	Employment	Humanitarian	Other
IRCA intensity × Post	−0.025***	−0.027**	−0.008	−0.016**	−0.009
	(0.005)	(0.011)	(0.011)	(0.006)	(0.009)
Adjustment share	−0.149*	−0.400**	−0.206	0.192**	0.108
	(0.081)	(0.152)	(0.128)	(0.091)	(0.138)
Num.Obs.	351	352	352	352	352
Mean of Y	0.619	0.542	0.301	0.067	0.013

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp per one-unit exposure. Net specs include adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Key estimate (Net, blue-collar share): Exposure × Post = -0.036*** (SE 0.007). Interpreted descriptively, a one-unit higher exposure is associated with a 3.6 pp lower blue-collar share during 1989–1992, relative to a mean blue share of 0.342. Other occupational shifts include higher agricultural share (+0.029) and lower "no occupation" share (-0.025), alongside changes in family and humanitarian admission shares (exploratory).

Sensitivity Checks

The blue-share association is stable across country trends, weights, balanced panel restrictions, excluding Mexico, and leave-one-country-out influence analysis.

Table 7: Extra sensitivity checks: dropping high-missing years, dropping incomplete countries, and 1982–1995 window (blue share, Net).

	Baseline Net	Drop 1995–1997	Drop UY+PY	Window 1982–1995
IRCA × Post	−0.036***	−0.042***	−0.037***	−0.037***
	(0.007)	(0.008)	(0.007)	(0.008)
Adjustment share	0.169*	0.224**	0.190*	0.182*
	(0.085)	(0.092)	(0.090)	(0.092)
Num.Obs.	351	300	323	260

* p <0.1, ** p <0.05, *** p <0.01
All models: Country & Year FE; SE clustered by country. Outcome: blue_share.

Table 8: Robustness to country-specific linear trends (blue share).

	Baseline Total	Baseline Net	Trends Total	Trends Net
IRCA intensity × Post	−0.025***	−0.036***	−0.025***	−0.036***
	(0.005)	(0.007)	(0.005)	(0.008)
Adjustment share		0.169*		0.160
		(0.085)		(0.099)
Num.Obs.	353	351	353	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Trends specs include origin-specific linear trends (country[t]).

Table 9: Specification comparison: baseline, trends, weights, balanced panel (blue share, Net).

	Baseline Net	Country trends	WLS admissions	Balanced Net
IRCA × Post	−0.036***	−0.036***	−0.036***	−0.037***
	(0.007)	(0.008)	(0.003)	(0.007)
Adjustment share	0.169*	0.160	0.150**	0.190*
	(0.085)	(0.099)	(0.069)	(0.090)
Num.Obs.	351	351	351	323

* p <0.1, ** p <0.05, *** p <0.01
Comparison of key robustness specifications for blue-collar share.

Table 10: Balanced panel restriction (blue share, Net).

	Blue Net (balanced)
IRCA × Post	−0.037***
	(0.007)
Adjustment share	0.190*
	(0.090)
Num.Obs.	323

* p <0.1, ** p <0.05, *** p <0.01
Balanced panel: keeps only countries observed in all years of the Net sample.

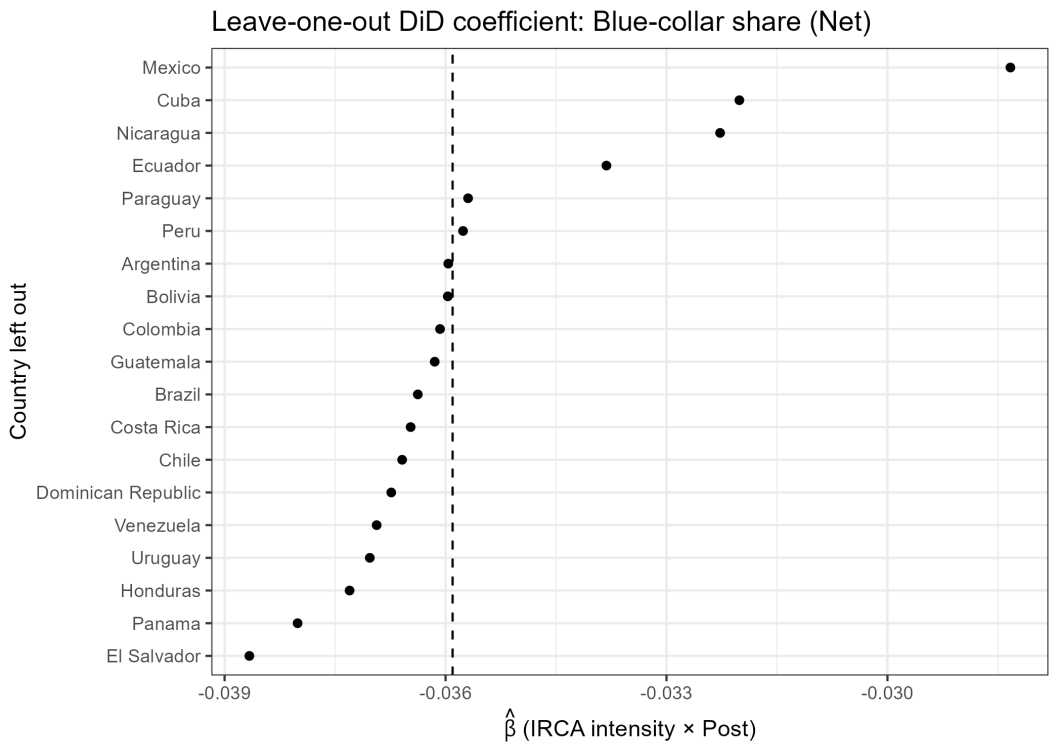

Figure 1: Leave-one-country-out sensitivity: estimated IRCA × Post coefficient (blue share, Net).

The leave-one-out plot is an influence diagnostic: it checks whether the headline association is driven by any single origin country. The coefficient remains negative and stays in a narrow band around the baseline when each country is dropped in turn.

Denominator Sensitivity and Decomposition

A key limitation is that results depend on how occupational shares are defined when there is a large "no occupation" mass.

- Using the Net denominator (known occupations only), the association is -0.036*** (SE 0.007).
- Using the Full denominator (known + unknown/no occupation), the association attenuates

to ≈ -0.005 (not significant).

Table 11: Denominator sensitivity: blue share using net (known occupations) vs full (known + no occupation) denominators.

	Blue (Net denom)	Blue (Full denom)
IRCA intensity × Post	−0.036***	−0.005
	(0.007)	(0.005)
Adjustment share	0.169*	0.127**
	(0.085)	(0.059)
Num.Obs.	351	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Full denom divides by known+unknown occupations.

Table 12: IRCA × Post associations on the no-occupation share (Net sample).

	No-occ share (Net sample)
IRCA × Post	−0.025***
	(0.005)
Adjustment share	−0.149*
	(0.081)
Num.Obs.	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Outcome: noocc_share.

To understand the mechanics behind a share result, I decompose the association into numerator and denominator components (levels and logs):

- Blue-collar counts increase with exposure in the post period (+9,683***).
- Known-occupation and total-occupation counts increase much more (+39k to +43k*).

Table 13: Decomposition — Panel A: levels (counts).

	Blue level	Occ known level	Occ total level	No-occ level
IRCA × Post	9682.575***	39 226.652***	43 404.916***	4178.264***
	(2342.817)	(9956.342)	(10 418.047)	(561.725)
Adjustment share	−973.413	−18 202.159	−204.179	17 997.980
	(5699.283)	(21 227.495)	(32 984.662)	(17 559.005)
Num.Obs.	352	352	352	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Levels are counts.

Table 14: Decomposition — Panel B: log(1+x) outcomes.

	ln(1+Blue)	ln(1+Occ known)	ln(1+Occ total)	ln(1+No-occ)
IRCA × Post	0.098*	0.192***	0.131***	0.056
	(0.051)	(0.037)	(0.036)	(0.034)
Adjustment share	1.092	0.776	0.608	0.224
	(0.636)	(0.469)	(0.470)	(0.513)
Num.Obs.	352	352	352	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Log specs use ln(1+x).

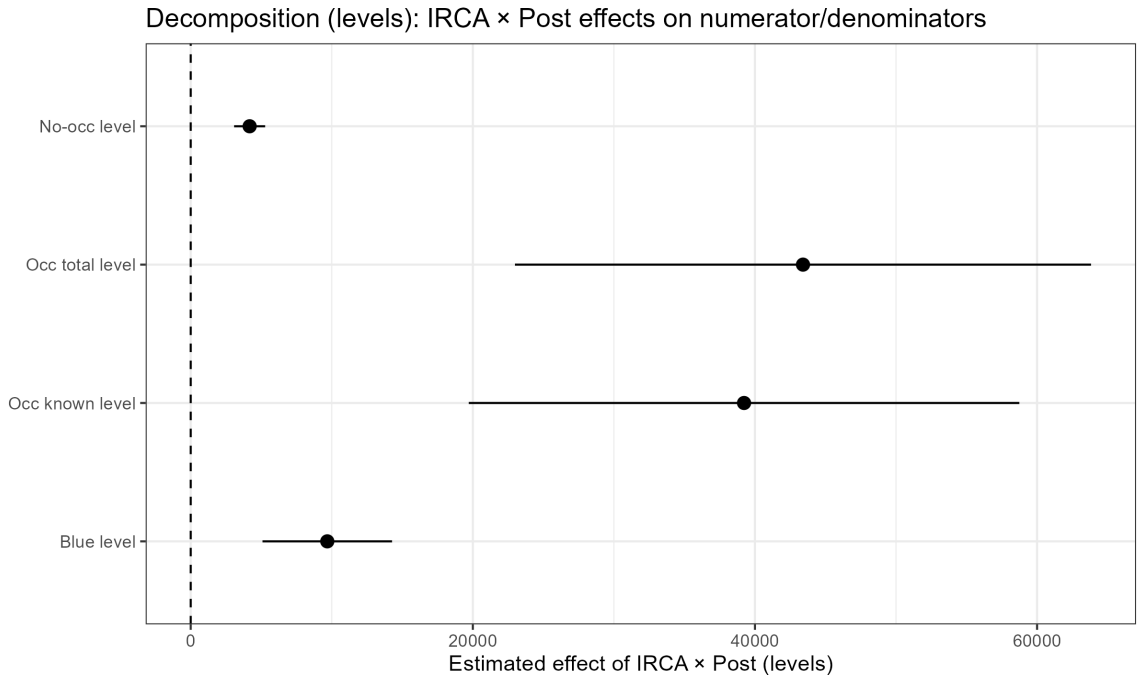

Figure 2: Decomposition (levels): IRCA × Post associations on blue counts and denominators.

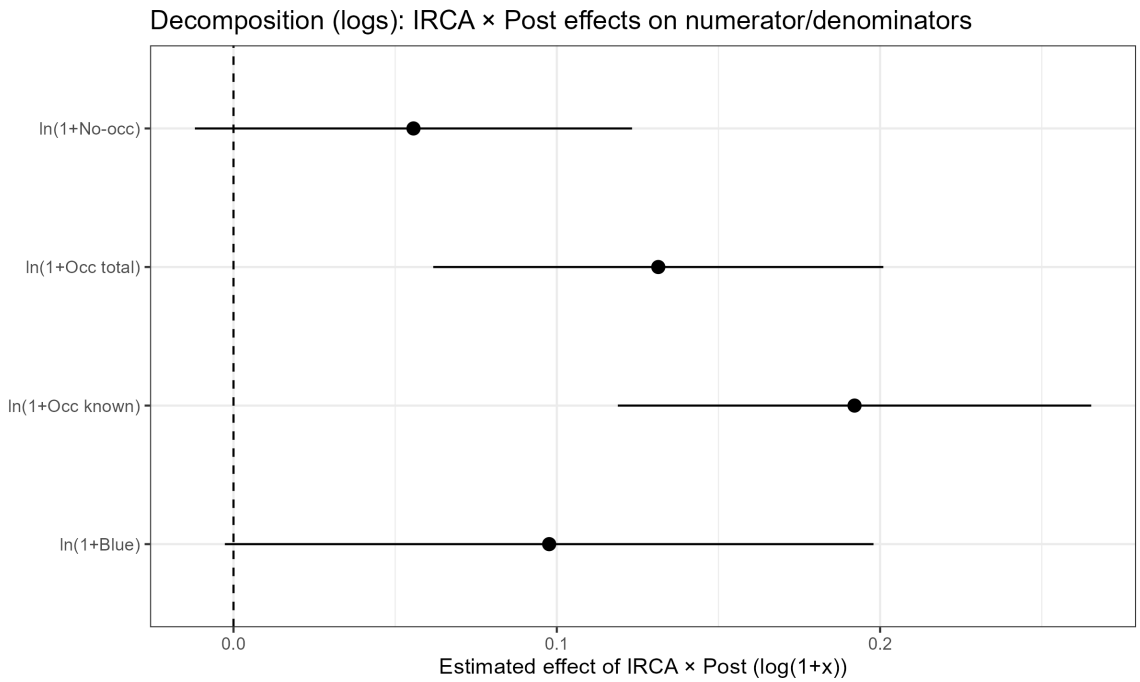

Figure 3: Decomposition (log(1+x)): IRCA × Post associations in proportional terms.

Overall, the blue-collar share decline reflects denominator growth outpacing numerator growth, rather than a contraction of blue-collar admissions in levels.

Economic Magnitudes

Exposure varies across origins (mean 1.476, SD 2.119). Using the baseline coefficient (-0.036), a 1 SD difference in exposure corresponds to:

$$\Delta Y \;=\; \beta \cdot SD(\text{Exposure}) \;=\; (-0.036)(2.119) \;=\; -0.0763 \;\Rightarrow\; -7.63 \text{ pp.}$$

$$\text{As \% of mean}(Y): \quad \frac{\Delta Y}{\bar{Y}} \times 100 \;=\; \frac{-0.0763}{0.342} \times 100 \;\approx\; -22.3\%.$$

These magnitudes should be interpreted alongside denominator sensitivity and timing diagnostics.

Table 15: Standardized magnitudes for the baseline Net specification (blue share).

Magnitude	Effect (pp / %)
Per 1 unit exposure	-3.59
Per 1 SD exposure	-7.61
Per IQR exposure	-2.46
As % of mean(Y): 1 SD	-22.22
As % of mean(Y): IQR	-7.18

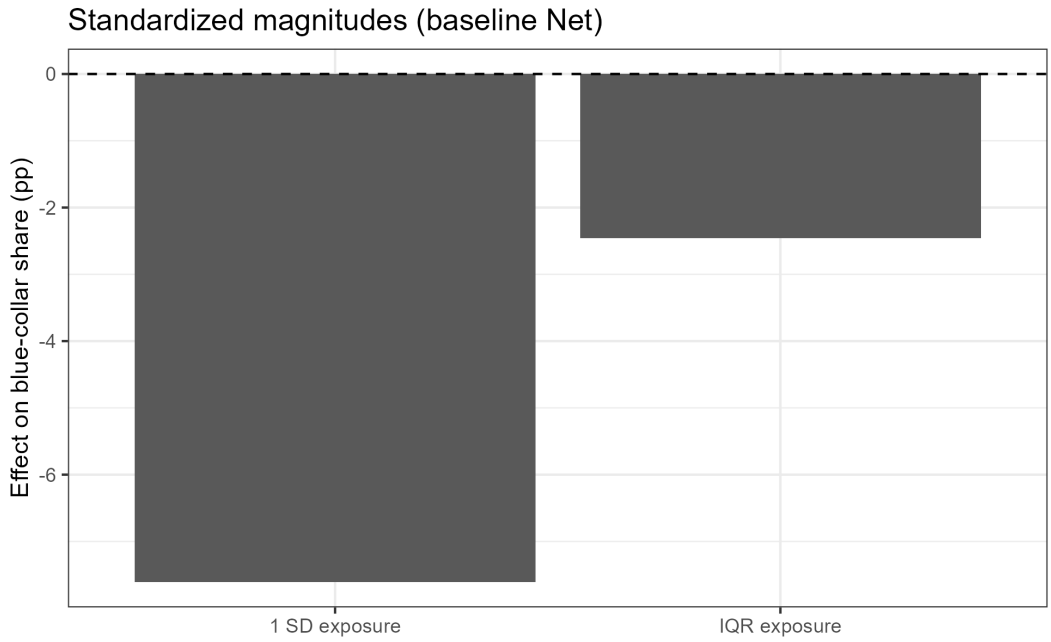

Figure 4: Standardized association sizes (1 SD and IQR exposure) for the baseline Net specification.

Timing Diagnostics (Interpretation Check)

These checks help interpret the timing of the exposure-gradient patterns. They are not used to claim causal effects.

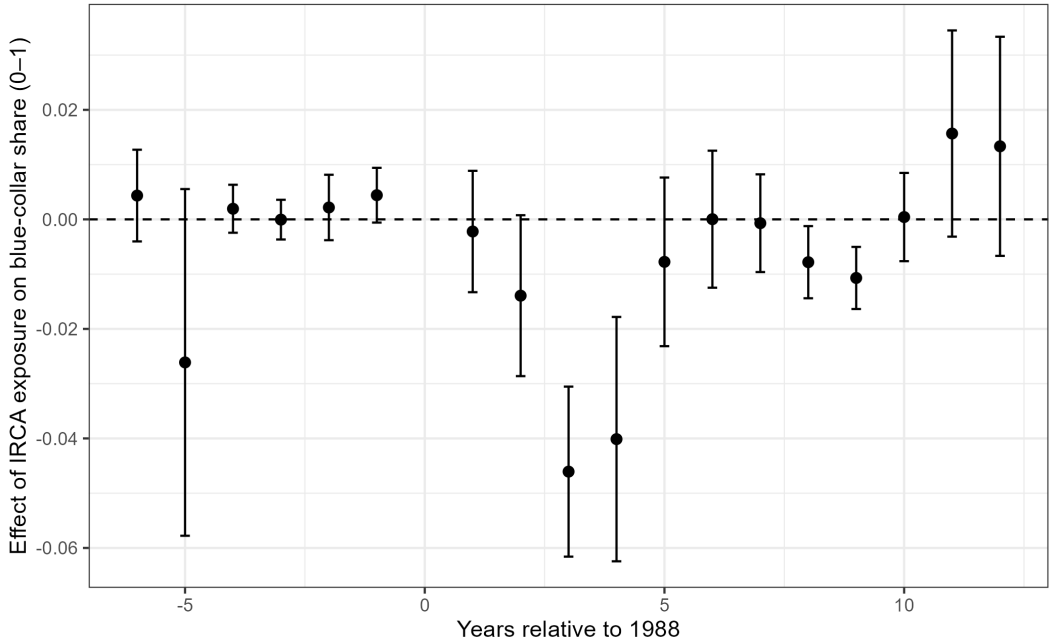

Figure 5: Event-study: exposure-by-year associations for blue-collar share (relative to 1988).

Formal pre-period regressions show no significant exposure-driven linear trends before 1989, but event-study pre-coefficients are jointly significant; I interpret dynamic estimates as descriptive and rely on sensitivity checks.

Table 16: Linear pretrend test (pre-period only): exposure × time slope.

	(1)
t	0.004
	(0.005)
irca_intensity × t	0.002
	(0.002)
Num.Obs.	132
R2	0.707
R2 Adj.	0.654
R2 Within	0.040
R2 Within Adj.	0.023
AIC	−276.7
BIC	−216.2
RMSE	0.07
Std.Errors	by: country
FE: country	X

* p <0.1, ** p <0.05, *** p <0.01
Pre-period only (<=1988). Country FE; SE clustered by country. Tests whether exposure predicts pre-trends.

Table 17: Placebo tests (pre-period only): fake post dates 1986 and 1987.

	Fake post >= 1986	Fake post >= 1987
Exposure × FakePost(86)	0.007	
	(0.006)	
Exposure × FakePost(87)		0.006
		(0.004)
Num.Obs.	132	132

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Pre-period only.

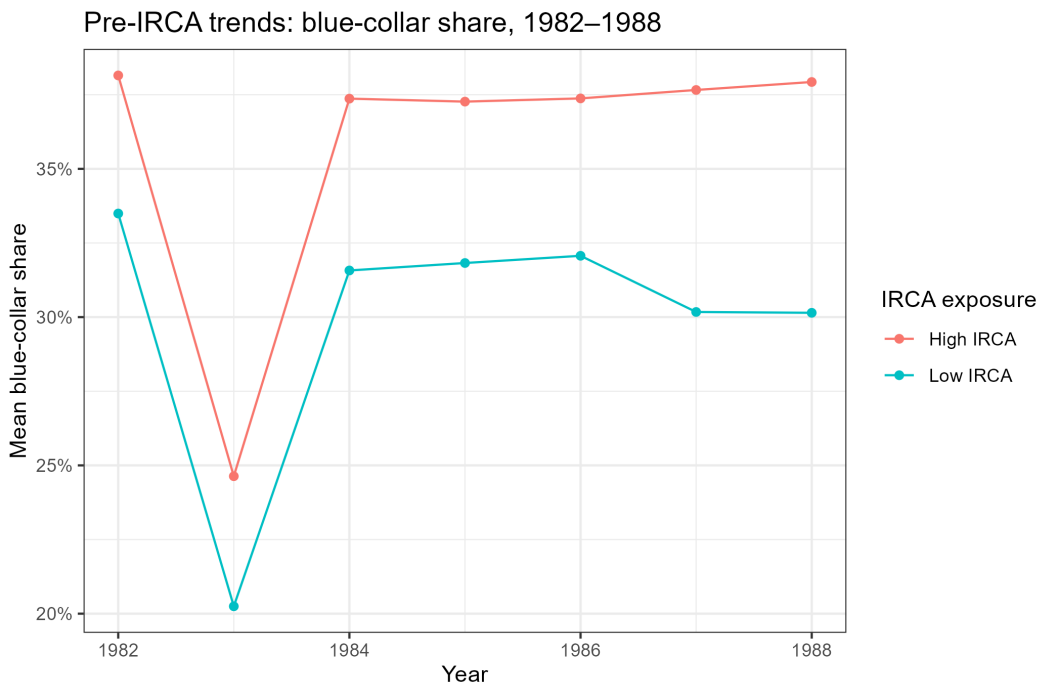

Figure 6: Pre-IRCA group means (1982–1988): blue-collar share by exposure group.

Heterogeneity by Exposure Level

Exposure quartiles reveal non-linear associations.

Table 18: Heterogeneity by exposure quartile: post-period associations with blue share (Net).

	Blue (Net) bins
Post × Q2	−0.006
	(0.016)
Post × Q3	−0.022
	(0.022)
Post × Q4	−0.134**
	(0.049)
Adjustment share	0.103
	(0.085)
Num.Obs.	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Exposure quartiles at country level; Q1 omitted.

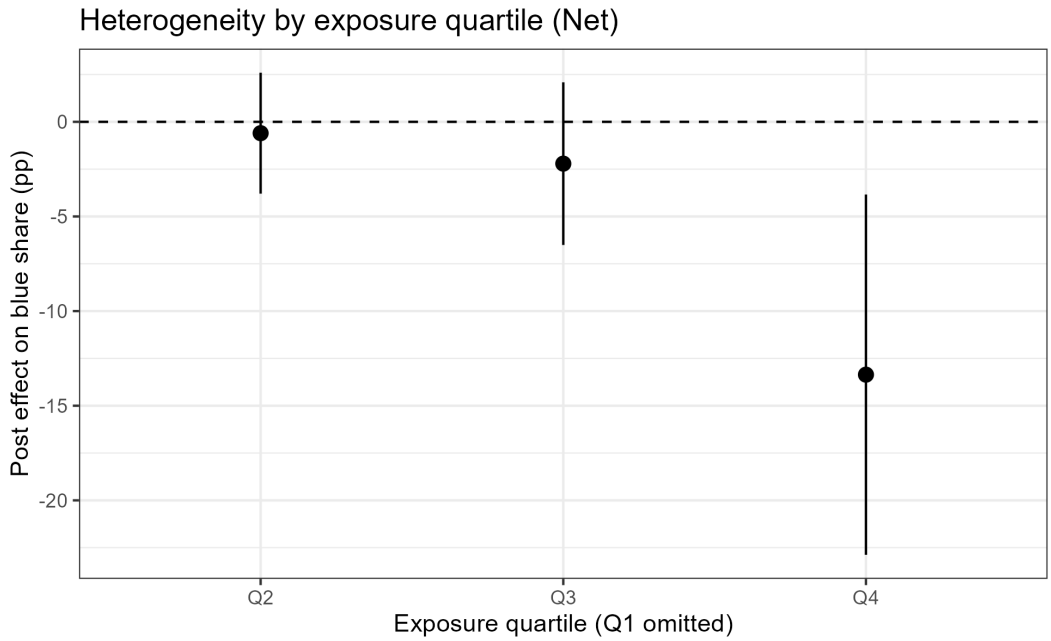

Figure 7: Heterogeneity by exposure quartile (Net): post associations on blue share with 95% CI.

Mechanism Exploration

I examine which dimensions co-move with exposure during 1989–1992 (exploratory; interpretation depends on data availability by outcome).

Table 19: Mechanism scan (exploratory) — Panel A: core outcomes.

	blue_share	serv_share	white_share	fam_share	emp_share
IRCA × Post	−0.036***	−0.006*	0.013*	−0.027**	−0.008
	(0.007)	(0.003)	(0.006)	(0.011)	(0.011)
Adjustment share	0.169*	−0.019	−0.144	−0.400**	−0.206
	(0.085)	(0.032)	(0.093)	(0.152)	(0.128)
Num.Obs.	351	351	351	352	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Panel A: core outcomes.

Table 20: Mechanism scan (exploratory) — Panel B: secondary outcomes.

	male_share	adj_share
IRCA × Post	0.017***	0.064***
	(0.003)	(0.010)
Adjustment share	−0.061	
	(0.061)	
Num.Obs.	133	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country.
Panel B: secondary outcomes (smaller samples for some outcomes).

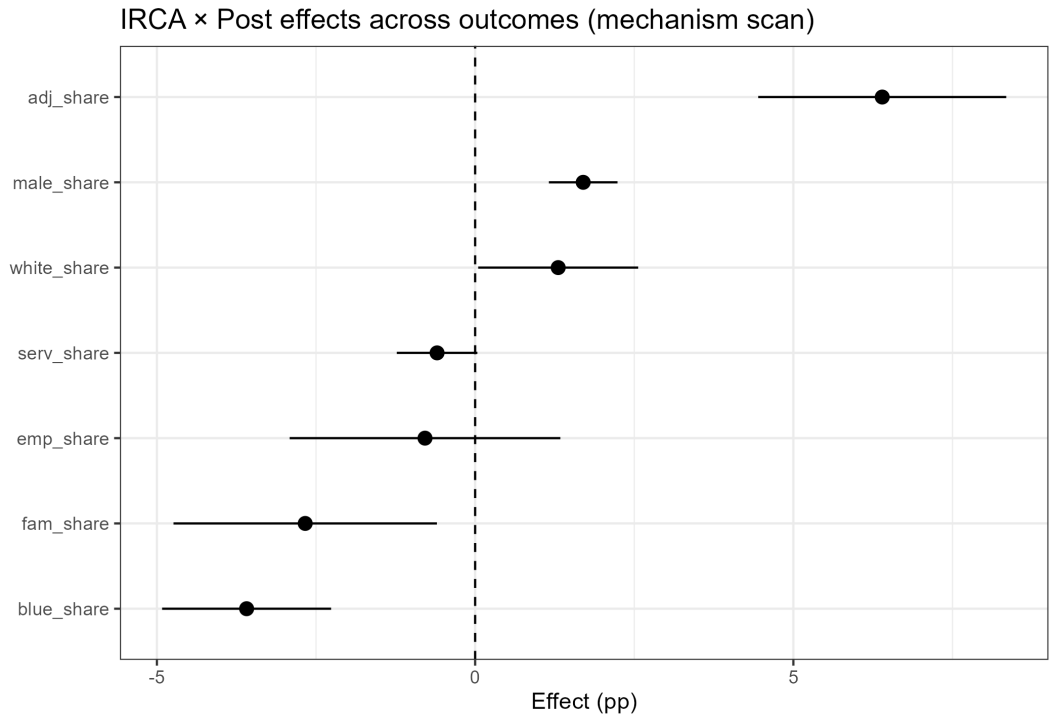

Figure 8: Mechanism scan: Exposure × Post associations across outcomes (exploratory).

These coefficients help prioritize monitoring and follow-up checks (e.g., if adjustment share co-moves with exposure, validate whether processing capacity or classification practices changed during 1989–1992).

Limitations

- Exposure is time-invariant by construction; inference relies on cross-sectional intensity differences plus fixed effects.
- Timing diagnostics are mixed (event-study pre-coefficients jointly significant), so dynamic interpretation should be cautious.

- Occupational coding contains a large unknown/no-occupation mass; denominator choice materially changes estimates.
- Cluster count is small (19 origins); the paper complements inference with sensitivity and influence diagnostics.

These are acknowledged as areas for future refinement.

Reproducibility

To reproduce results end-to-end: run Rscript run.R and then render paper.Rmd.

Outputs:

- Tables → outputs/tables/
- Figures → outputs/figures/
- Logs → outputs/logs/
- Clean panel → data_clean/panel.rds

Appendix: Residual sanity check

The residual-by-year plot is a basic sanity check (mean residuals are mechanically near zero under fixed effects) and is included only as a QA artifact.

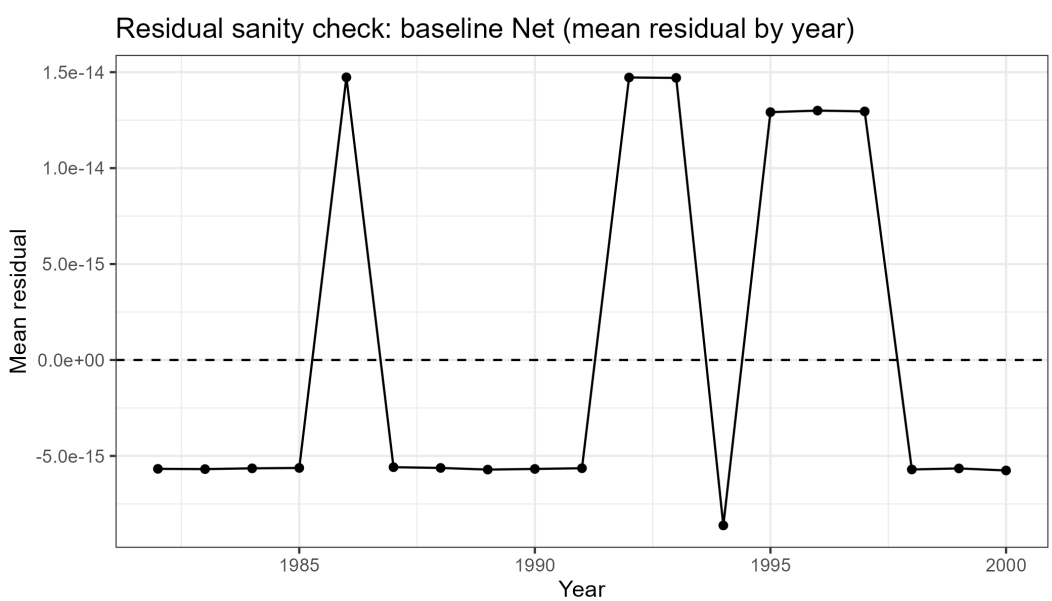

Figure 9: Residual sanity check: mean residual by year (baseline Net, blue share).